

LAB REPORT 1 [LAB1]
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3-02-2025

ABSTRACT. This laboratory focuses on the use of Matlab in experimenting plotting, vectors, and digital image scalar fields.

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1. PROBLEMS

lab1-1 Modify the `exmple1LinewidthProb1.m` Matlab script from Lecture 1 to obtain the required plot.

The solution is shown in Figure 1 below and the complete code is available in the appendix A.0.1

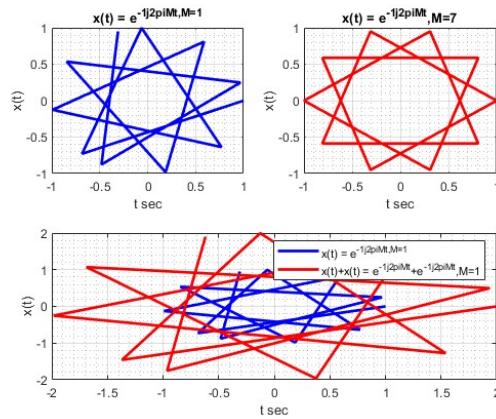


FIGURE 1. Solution visualization for the modified MATLAB script problem 1.

Key words and phrases. Euler Exponential, Frequency (via Mersenne Primes), Matlab, Pixel, Polar Form, Scalar Field, Digital Image, Vector.

lab1-2 Modify the `exmpleExTanhProb2.m` Matlab script from Lecture 1 to obtain the required plot.

The solution is shown in Figure 2 below. The code is in the Appendix A.0.2

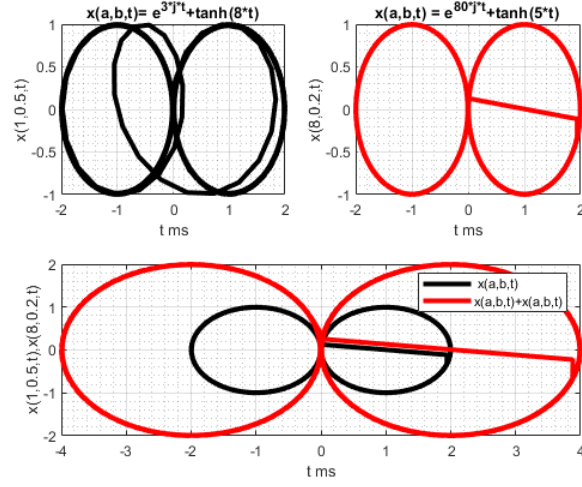


FIGURE 2. Solution visualization for the modified MATLAB script problem 2.

lab1-3 Modify `exmpleEulerProb3.m` Matlab script from Lecture 1 to obtain the required plot. The solution is shown in Figure 3 below. The code is in the Appendix A.0.3

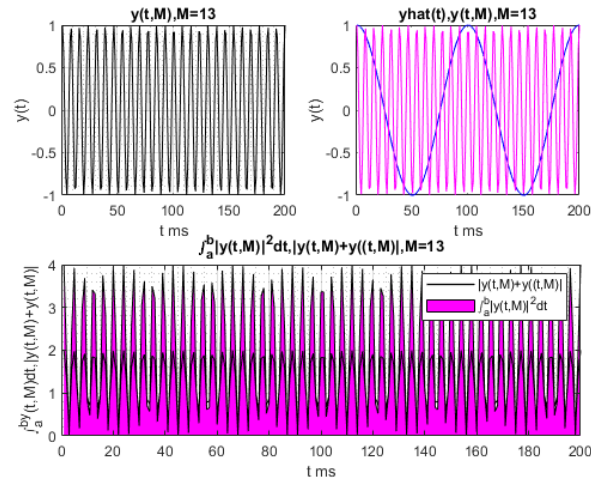


FIGURE 3. Solution visualization for the modified MATLAB script problem 3.

lab1-4 Modify the testImgScalarField.m Matlab script from Lecture 1 to obtain the 3 required subplots. The solution is shown in Figure 4 below. The code is in the Appendix A.0.4

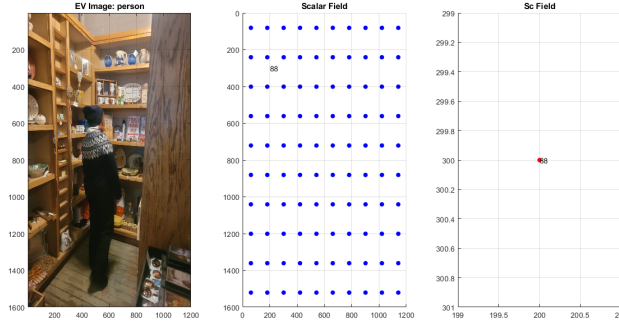


FIGURE 4. Solution visualization for the modified MATLAB script problem 4.

lab1-5 Prove:

Theorem 1. *Every digital image is a scalar field.*

To start proving the theorem first we need to have 2 main definitions, first a **Scalar** is a value that has a point, it can be for example in a black and white image a number from 0 to 255. Second **Scalar Field**, it is a function that assigns a single scalar to every point in a space (in any dimension). For example, temperature across a room is a scalar field: each point (x, y) has a temperature value (scalar). Finally, a **Digital Image** is a grid of small squares called pixels, where each pixel has a position (x, y) and a value representing its brightness or color.

Now consider a digital black and white image X and set a coordinate system to it where the lower-left corner of it is the origin $(0, 0)$. Each pixel in the image has coordinates (x, y) , where:

- x increases to the right (horizontal axis),
- y increases upward (vertical axis).

With the previous we can start, let's define a function $f(x, y)$ that gives the intensity (brightness) of the pixel at position (x, y) :

$$f : \mathbb{Z} \rightarrow \mathbb{R}, \quad f(x, y) = I(x, y)$$

In the function $\mathbb{Z}$ is the set of integer pixel coordinates (x, y) and $I(x, y)$ is the intensity value (e.g., 0 for black, 255 for white in grayscale). This mathematical representation makes the function f assign a scalar $I(x, y)$ to each discrete coordinate (pixel) (x, y) , in this example is a grayscale picture, for color images, three scalar fields exist—one for each red, green, and blue.

lab1-6 Prove:

Theorem 2. *Every pixel in a digital image is a vector.*

First, we have to define the structure of a digital image, it consists of pixels arranged on a grid. To prove that every pixel is a vector, we first remember that a vector is defined by both magnitude and direction, as a set of components that can be added or scaled.

Consider the lower-left corner of the image as the origin $(0,0)$. Each pixel has coordinates (x, y) , where x increases horizontally to the right and y increases vertically upward. The position of a pixel can be represented as a **position vector**:

$$\mathbf{v} = x \mathbf{i} + y \mathbf{j},$$

Where $\mathbf{i}$ and $\mathbf{j}$ are unit vectors. This vector has a magnitude (distance from the origin) given by $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$ and a direction θ relative to the axis x . Thanks to that, the pixel's position inherently satisfies the definition of a vector.

lab1-7 In an Observations section of your report, indicate anything special or interesting that you found in solving one or more of the lab problems.

I found it interesting how playing with the `linespace` in Matlab, the function can generate so many different solutions, no matter if is the same jump in time, the graph `-1:0.23:1` is different from `-2:0.23:2`, and also how the size of the figure changes in the point 4 changing the size of the image.

2. BIOGRAPHY

I was born in Bogotá, Colombia, both of my parents were studying engineering when I was born. I grew up in the same city and, after finishing high school, I pursued a Bachelor's degree in Electronic Engineering at Javeriana University, one of the best universities in my country. During my studies, I met my fiancée, which has been one of the most important experiences of my life.



FIGURE 5. My photo

Since I was a child, I've been passionate about three things: photography, technology, and nature. This combination of interests led my career, as I hope to use engineering to benefit the natural world. Photography is my main hobby, but I also love trying new experiences. That curiosity helped me to gain a diving certification. I'm currently pursuing a Master of Engineering at the University of Manitoba. While I'm not working on a research project, I'm interested in image processing. One of the topics in this course that excites me the most is understanding how computers interpret images from scratch. I hope to apply this knowledge in my final project, possibly integrating it with my passion for photography, technology, and nature.

APPENDIX A. SAMPLE MATLAB SCRIPT

```

A.0.1. Lab1-1.m
% Lab1-1
clc, clear all, close all; % housekeeping
syms t; % vector t
x = @(t) exp(-1*i*2*pi*t); % define function x(t)
x2 = @(t2) exp(-1*i*2*pi*7*t2); % define function x(t)

% start using x(t)
t = -2:0.37:+2; % set up vector t
t2 = -2:0.1:+2; % set up vector t
subplot(2,2,1), % set up row of 2 plots
plot(x(t), 'b', 'LineWidth', 2, 'MarkerSize', 10),
xlabel('t sec'), ylabel('x(t)'); % insert labels
title('x(t) = e^{-1j2piMt}, M=1'); % set up plot title
grid on, grid minor;

subplot(2,2,2), % 2nd row plot
plot(x2(t2), 'r', 'LineWidth', 2, 'MarkerSize', 10),
xlabel('t sec'), ylabel('x(t)');
title('x(t) = e^{-1j2piMt}, M=7'); % set up plot title
grid on, grid minor;

subplot(2,2,[3,4]),
plot(x(t), 'b', 'LineWidth', 2, 'MarkerSize', 10),
hold on;
plot(x(t)+x(t), 'r', 'LineWidth', 2, 'MarkerSize', 10),
xlabel('t sec'), ylabel('x(t)'); % insert labels
grid on, grid minor;
xlabel('t sec'), ylabel('x(t)');
legend('x(t) = e^{-1j2piMt}, M=1', 'x(t)+x(t) = e^{-1j2piMt}+e^{-1j2piMt}, M=1'); % set up legend

```

```

A.0.2. Lab1-2.
%Lab1-2
clc, clear all, close all;

syms a b t; % introduce 3 variables
x = @(a,b,t) exp(b.*1i*t)+tanh(-a.*t); % introduce new fn}

t = linspace(-8,+2*pi,300); % small sample

subplot(2,2,1), a = 3;b = 8;
plot(-x(a,b,t),'k','LineWidth',3,...
     'MarkerSize',5), % discrete fn values
grid, grid minor,
xlabel('t ms'), ylabel('x(1,0.5,t)');
title('x(a,b,t)= e^{3*j*t}+tanh(8*t)')

subplot(2,2,2), a = 80;b = 5;
plot(x(a,b,t),'r','LineWidth',3,...
     'MarkerSize',2), grid, grid minor,
xlabel('t ms'), ylabel('x(8,0.2,t)');
title('x(a,b,t) = e^{80*j*t}+tanh(5*t)')

subplot(2,2,[3,4])
plot(x(a,b,t),'k','LineWidth',3,...
     'MarkerSize',5), hold on,
plot(x(a,b,t)+x(a,b,t),'r','LineWidth',3,...
     'MarkerSize',10),hold off;grid, grid minor,
xlabel('t ms'), ylabel('x(1,0.5,t),x(8,0.2,t)');
legend(' x(a,b,t)', 'x(a,b,t)+x(a,b,t)')

```

```

A.0.3. Lab1-3.
% Lab1-3

clc, clear all, close all;

syms t t2 freq;

y = @(t,freq) real(cos(freq.*t)-1i.*sin(freq.*t)); % Euler

yhat = @(t2) real(cos(t2)-1i.*sin(t2));

t = linspace(-2*pi,+2*pi,200);
t2 = linspace(-2*pi,+2*pi,200);
freq = 13;
subplot(2,2,1),
plot(y(t,freq),'k','LineWidth',1,...
     'MarkerSize',10), grid, grid minor,
xlabel('t ms'), ylabel('y(t)');
title('y(t,M),M=13')

subplot(2,2,2),
plot(yhat(t2),'b','LineWidth',1,...
     'MarkerSize',10), grid, grid minor,
xlabel('t ms'), ylabel('yhat(t)');
hold on
plot(y(t,freq),'m','LineWidth',1,...
     'MarkerSize',10), grid, grid minor
xlabel('t ms'), ylabel('y(t)');
hold off
title('yhat(t),y(t,M),M=13')

subplot(2,2,[3,4]),
area(abs(y(t,freq)+y(t,freq)).^2,'FaceColor','m'),
hold on,
plot(abs(y(t,freq)+y(t,freq)),'-k','LineWidth',1,...
     'MarkerSize',2),
hold off,
abs(integral2(y,0,200,1,13))
grid, grid minor,
xlabel('t ms'), ylabel(['\int_{a}^{b}' ...
     '\tilde{y}(t,M)dt, |y(t,M)+y(t,M)|']);
legend('\int_{a}^{b}|y(t,M)+y((t,M)|^2dt', '|y(t,M)+y((t,M)|');
title('\int_{a}^{b}|y(t,M)|^2dt, |y(t,M)+y((t,M)|,M=13')

```

A.0.4. Lab1-4.

```

% Lab1-4

clc, clear, close all;

%Initialization
EXP_NAME='_ImgScalarField';
Img_name = ['person'];
IMG0 = imread([Img_name, '.jpeg']);
IMG = rgb2gray(IMG0);
[row,col] = size(IMG);

%calculation
x = round(row/20)+1:round(row/10):round(row/20)+row;
y = round(col/20)+1:round(col/10):round(col/20)+col;
[xx,yy] = meshgrid(y,x); %construct the vector field using motions

%visualization
f = figure;

subplot(1,3,1),% plot the frame
imagesc(IMG0),grid on,axis on;
title(['EV Image: ',Img_name]);

subplot(1,3,2); %plot the vector field
scatter(xx,yy,'o','b','filled');
text(200,300,'88')
grid on,axis on;
set(gca, 'YDir','reverse');
title('Scalar Field')

subplot(1,3,3); %plot the vector
scatter(200,300,'o','r','filled');
text(200,300,'88')
grid on,axis on;
set(gca, 'YDir','reverse');
title('Sc Field')

disp('finish')

```